Cyber Power as a Structural Variable of Maritime Power

A conceptual update of its formulation

EXTENDED MARITIME POWER FORMULA

$$MP = \left[\left(MI \times PN \right) \times SW \times MS^{\gamma} \times CP^{\alpha} \right]^{1/n_{eff}}$$

$n_{eff} = 3+\gamma+\alpha$ · $MP_{norm} \in (0,1]$ · Normalization: geometric mean (UNDP 2010; OECD 2008)

ORIGINAL CONTRIBUTIONS OF THE ARTICLE

- **Cyber power (CP)** as a structural variable of maritime power — first formalization in the Spanish-language literature.
- **Strategic will (SW)** repositioned as a multiplier of the complete MP system, not only of naval power.
- **Maritime security (MS)** disaggregated into three dimensions: interstate conflicts, terrorism, and blue crime.
- **Normalization via n_{eff}** → $MP_{norm} \in (0,1]$ with a directly interpretable five-level scale.
- **CP is the most efficient investment vector** for middle powers ($\alpha < 1$): larger gap, high initial returns, transversal enabling character.

5	2	5	3
structural variables formalized	disaggregation levels (macro + 27 sub-indicators)	levels of the analytical scale (0,1]	verification stages

COLOMBIA — MP PROFILE *ref: Vargas Suárez et al., 2021*

0.53

★ MODERATE LEVEL

MI	NP	SW	MS	CP
0.60	0.55	0.48	0.59	0.40

COMMON-SCALE ANCHORING · INDEX 0.44 (RANK 33/99)

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BASE

THE ESSENTIAL IDEA · CLASSICAL CORE

MI × **PN** → **MP**

Maritime Power = Maritime Interests × Naval Power

Point of departure: the classical formulation (Solís Oyarzún, 1997; Uribe Cáceres et al., 2016) expresses maritime power as the product of what the State wants at sea (interests) and its capacity to defend it (naval power). The multiplicative nature imposes that if either factor collapses, so does the result.

The ***classical formula*** *is the foundation this research extends: sound, but blind to the digital domain.*

EQ. 1b–1c **NORMALIZATION** · FROM RAW PRODUCT TO INDEX

$$MP_{norm} = [\ MP\ ]^{1/n_{eff}}$$

$$n_{eff} = 3 + \gamma + \alpha$$

Why normalize: multiplying five components in (0,1] produces very small values. The weighted geometric mean —following UNDP (2010) and OECD (2008)— rescales the result to the interval (0, 1], where a state with all components equal to 1 obtains exactly 1. The exponent n_{eff} equals the sum of all component exponents: 1 (MI) + 1 (NP) + 1 (SW) + γ (MS) + α (CP).

Normalization turns an abstract product into an ***interpretable index*** *on a five-level scale.*

EQ. 2 MARITIME INTERESTS · LEVEL 2

$$\boxed{MI} = \Sigma\, w_i \cdot MI_i$$

MI_i

i-th maritime interest

Each of the national maritime interests: resources, trade, sovereignty, science, environment.

w_i

Interest weight

Relative strategic priority assigned to each interest within the national set.

Σ

Aggregation

Weighted sum that integrates all interests into a single comparable sub-index.

Calibration: weights $w_i \in (0,1]$ sum to 1 ($\Sigma w_i = 1$). For Colombia, the CCO (2021) recognizes 18 national maritime interests.

EQ. 3 NAVAL POWER · LEVEL 2

$$NP = F \times P \times MDA^{\mu}$$

F

Force

Naval platforms, weapons systems, trained personnel, C2, and tactical cyber defense.

P

Position

Strategic geography, naval bases, area control, and access to maritime traffic routes.

MDA^{μ}

Maritime Domain Awareness

Operational awareness of the domain, elevated to a structural factor with calibration exponent μ.

Key decision: MDA is elevated to the level of F × P because, without domain awareness, force and position do not translate into effective naval power (Bebber, 2022).

EQ. 4 STRATEGIC WILL · LEVEL 2

$$SW = (PW^{w_1} \cdot IW^{w_2} \cdot SocW^{w_3})^{1/\Sigma w}$$

PW
Political Will

Decision of state leaders to prioritize and sustain national maritime interests.

IW
Institutional Will

Capacity of the institutional apparatus to translate decisions into norms and structures.

SocW
Social Will

Capacity of society to turn maritime consciousness into appropriation and effective use.

Contribution: strategic will operates as a multiplier of the complete system (not only of naval power). The geometric mean prevents any dimension from compensating a collapsed one.

EQ. 5 MARITIME SECURITY · LEVEL 2

$$MS = (IC^{a_1} \cdot TE^{a_2} \cdot BC^{a_3})^{1/\Sigma a}$$

IC
Interstate Conflicts

Border and maritime resource disputes, plus gray-zone coercive activities below the armed-conflict threshold.

TE
Terrorism & Extremism

Attacks against vessels, platforms, and port facilities; the sea as a logistical corridor.

BC
Blue Crime

Piracy, drug and human trafficking, and environmental crimes (IUU fishing, pollution).

Framework: the three dimensions follow the maritime security agenda of Bueger & Edmunds (2024). The global exponent γ weights the environment over total maritime power.

EQ. 6 **CYBER POWER** · CENTRAL CONTRIBUTION

$$CP = \min\{1, (\beta_1 \cdot CS_m + \beta_2 \cdot CD_m + \beta_3 \cdot CR_m) \cdot (1 + \delta \cdot CDip)\}$$

CS_m
Cyber Security

Protection of the digital assets of the maritime domain: AIS, ECDIS, GMDSS, port infrastructure.

CD_m
Cyber Defense

Strategic national capacity to operate and defend naval cyberspace at policy and doctrine level.

CR_m
Cyber Resilience

Operational continuity in the face of cyber incidents: detection, containment, recovery, adaptation.

CDip
Cyber Diplomacy

International projection through UN and OEWG normative frameworks. Amplifying factor.

Concavity: the exponent α (SCMI) < 1 makes CP^{α} concave — its marginal returns are highest at low cyber maturity, which makes CP the most efficient investment vector for middle powers.

EQ. 7 FULLY EXPANDED FORMULA · THE TWO LEVELS IN ONE EXPRESSION

$$MP_{norm} = \{ [(\Sigma\, w_i \cdot MI_i) \times (F \cdot P \cdot MDA^{\mu})] \times SW \times MS^{\gamma} \times CP^{\alpha} \}^{1/(3+\gamma+\alpha)}$$

The complete model: substituting each Level-2 component (Eqs. 2–6) into the normalized expression (Eq. 1b) yields the fully expanded formula. Every parameter is calibratable by state without altering the fundamental architecture, and the output always satisfies $MP_{norm} \in (0, 1]$, interpretable on the five-level scale.

From the **essential idea** *to the* **complete expression***: five structural variables, one interpretable index.*